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**Project Description:** Dynamic Light and Shadow Simulation to model a campfire scene with rain, Tung Tung Tung Sahur (ball knowledge required), trees, dynamic rain, and a moon and sun that render and are animated through the sky depending on the time of day. We used different lighting types (directional, point, spot, area), dynamic shadows, soft shadows, ambient occlusion, and time of day lighting changes.

**Steps Taken:** First and foremost, we had to set up the scene to provide a base for all the features and objects we were going to add. We made the ground and camp area in our function createTerrain(), using plane geometry for the ground and circle geometry for the camp area to differentiate it from the rest of the ground it sits on top of. We also created a moon and a sun, where the sun passes above and through the scene during the day in an orbit fashion and uses directional lighting to illuminate the scene, while the moon renders after the sun in the night and does the same thing. After that, we used createForest() to create trees that are scattered along our scene. The geometry used for the trees is very simple, utilizing a brown cylinder for the trunk and a green cone on top to represent the leaves on top. The main purpose of this forest is to show how shadows are produced in our scene - as the sun passes overhead through the day, we see our shadows starting long and being the shortest at noon, and then stretching in the opposite direction towards the evening before disappearing when the sun sets. After that, we wrote 2 methods to create a lake in the scene. We have a function that creates an outline for the lake, and another function that uses the outline function to create the lake with its properties. The lake also handles reflections, so we can see the reflections of the sun and moon as it passes over the lake. Next, we added a campfire to the scene inside the circle area we created earlier, using spot, point, and area lighting. We use spot lighting to shine directly onto the center of our campfire to illuminate the campfire area, point lighting to make the campfire glow, and area lighting to illuminate the surrounding area around the campfire. We also added torches around the lake we created as another light source, like the campfire, but unlike the campfire, we primarily use area lighting to illuminate the surrounding area around the torches. We have a custom campfire shader file that creates custom shaders for the flame, sparks, and smoke, which are used to add detail to the campfire. We further wanted to enhance the detail of the scene, so we added 2 different kinds of wildlife - we added birds that are animated to fly overhead during the day, and fireflies that are rendered when the sun starts going down, which faintly glow. We created a shader file for wildlife that handles the vertex and fragment shaders of the fireflies, where we give each firefly a pulse and a glow. For some extra creativity, we added a few features to make the scene a bit more memorable. On the left sidebar, we have several toggles for the user to choose from - many of these options control what we just discussed above, such as shadow mapping quality, soft shadows, the time of the day + how fast the day cycles, the campfire intensity, etc. However, there are a few more options we added, such as allowing the user to toggle rain on and off, choose if they want fog added to the scene, and wet ground that makes the ground a bit more shiny when rain is falling. We also have different lighting styles, such as realistic, cinematic, and stylized. We also added a walking Tung Tung Tung Sahur to the scene, where he paces around the campfire in a circular fashion while carrying his signature bat. In case the user wants a different view of the scene, we included an auto camera feature where the camera moves through a set of predefined points to give the appearance of it moving through the scene at a certain speed the user can set. For some extra detail to the scene, we added mountains that render in the distance to give some more depth to the scene. We also added mushrooms and flowers to the ground so the ground isn't bare bones, as well as a pier with a boat, the reflections of which appear in the lake. At nighttime, we also added an aurora in the sky for some extra lighting, as well as some shooting stars. Finally, we added a lone cabin that sits in the forest to give the scene some more detail, as well as a deer that walks around the forest and campground.

**Challenges:** Since this project was pretty complicated and we implemented a lot of extra features, we definitely faced some challenges when implementing them. For one, getting the camera to move through the scene in the auto camera feature was not very simple to do, since we did not want it moving in a linear fashion left to right or up to down. To address this, we stored a set of preexisting points that the camera loops through at a set pace, so we can control what exactly the user sees. Another issue was the different shadow map resolutions - we can let the user choose between low, medium, high, and ultra resolution shadows. However, getting this implemented was difficult since we didn't know what resolution to set each shadow setting at, which took a lot of trial and error to figure it out. One last issue we had was creating Tung Tung Sahur, since we wanted to depict him accurately and make him move around. We ended up making his body and limbs cylinders, along with the bat he holds, then using spheres to make his eyes, then set him to animate in a circular path around the campfire.

**Effort Justification:** This project represents at least a 2.5x effort of a regular homework assignment, as it's similar to other assignments we've had, but the scene itself is much more complicated than others we've made. In this project, we have a lot more moving parts - there are multiple different kinds of lighting implemented, different animated objects that must be conditionally toggled or rendered only at certain times of the day, multiple shader files, dynamic shadows that can have their resolution changed and change in length depending on the position of the sun, dynamic camera movements, reflections from the lake, and even a way for us to change the environment by using rain and fog. Each of us contributed 1.5x individually, as there was a lot of planning required which required both of us to collaborate/communicate efficiently, and each of us took certain features we wanted to see implemented and divided up the work. Overall, each of us still had to do more work than a single homework assignment to make this scene as complicated and customizable as we did.

**Rubric Checklist:**

| Points | Rubric Criteria |
|---|---|
| 25 | Implementation of different lighting types - includes directional, point, spot, and area lighting. Code should run without errors and lighting differences across modes should clearly be visible. |
| 15 | Implementation of dynamic shadows, soft shadows, and ambient occlusion. Code should run without errors and the shadows should truly be dynamic and not static. Shadows should be soft and not sharp/unrealistic, with the ability to change the shadow mapping resolution. |
| 15 | Extra effort for making the scene appear realistic - one example is making the scene change over time to simulate a full day, or adding multiple dynamic elements that allow the user to change how the scene appears. |
| 15 | Report + demo video |
| 70 | TOTAL |

1) 25 points - we implemented all different kinds of lighting (spot, point, directional, and area) in multiple different areas of our scene. The campfire uses spot, point, and area lighting, the torches use point lighting primarily, and the sun uses directional lighting to illuminate the scene as it passes by. The cabin windows and lanterns also use point light sources.
2) 15 points - we implemented dynamic shadows that change in length throughout the day (start long, get shorter towards noon, and get stretched out in the opposite direction towards the evening, similar to how shadows work in the real world). Shadows can be changed to be mapped to different resolutions, where the higher resolutions are the most computationally expensive, and the lower resolutions appear much fuzzier but improve performance. Soft shadows are also able to be toggled on or off. We also implemented ambient occlusion to make the scene appear less sharp and more realistic, which can be toggled on and off.
3) 15 points - the scene goes above and beyond to be realistic. We implemented the dynamic day and night cycle, where the sun passes by overhead during the day and when the sun sets, the moon renders and passes over the scene, with stars in the night sky using particles. The lake in the scene also captures reflections from the passing sun or moon depending on the time of the day. The scene also allows for dynamic weather by allowing us to toggle rain, wet ground, and fog, as well as setting the intensities for each type of weather feature. We also added wildlife, such as Tung Tung Tung Sahur, fireflies, the deer, and birds, which are all animated to interact with the scene. At night, we added an aurora and shooting stars to provide some extra light to the dark nighttime scene.

Work distribution for rubric:

Sanjay:

- Implemented sun in the scene.
- Added lake, boat, trees, mountains, and cabin to the scene.
- Added campfire to the scene with spot, point, and area lighting
- Added area lighting to the lake.
- Added point light sources to cabin windows.
- Added point light sources around the lake.
- Implemented a dynamic path for the sun to take throughout the day so it renders dynamic shadows that change in length depending on where the sun is using directional lighting.
- Added torches around the lake using point light sources.

Jayden:

- Implemented moon in the scene.
- Added aurora in the sky, stars using particle systems/point geometry, shooting stars, different wildlife (such as Tung Tung Tung Sahur, fireflies, flowers and mushrooms, deer).
- Added animation to wildlife in the scene.
- Added ambient occlusion and shadow resolutions.
- Added lanterns to the map using point light sources.
- Used particle system to make smoke above campfire to make it more realistic.
- Implemented shaders for campfire and wildlife.
- Implemented dynamic weather.

**Effort Justification (Sanjay):** this represents a 1.5x effort compared to a homework assignment on my end because of the level of depth of the scene I helped create. In previous homeworks, we would use different kinds of lighting, but the scene would largely be static with a moving light source like a sun to illuminate objects it passes by. However, in this scene I added multiple objects and light sources that demonstrate my full understanding of light and shadow. I added a sun to the scene using directional lighting so it dynamically adjusts the shadows of whatever it illuminates depending on the position, a lake that uses a reflector to reflect the surrounding scene , multiple point sources of light (such as the cabin windows and torches around the perimeter of the lake), and a campfire that utilizes 3 different kinds of lighting all at once to appear realistic (spot, point, and area). Getting these to work was not an easy task, as there are lots of moving parts and it was a challenge to make sure that all different kinds of lighting I used appeared as realistic as possible. Furthermore, the different light sources I used are not static - they dynamically adjust with the time of day, where light sources such as the cabin windows and fire are not as bright during the day but have a strong glow during the night.

**Effort Justification (Jayden):** This represents a 1.5x effort compared to a homework assignment on my end because a lot of the work I contributed added another layer of realism and technical complexity to the scene beyond just placing objects into it. In previous assignments, we worked with lighting and shadows in more controlled scenes, but for this project I helped implement features that made the environment feel dynamic and responsive. I implemented the moon and nighttime sky elements, but a major part of my contribution was focused on improving how realistic the scene looked through ambient occlusion, adjustable shadow resolutions, weather effects, and custom shaders. Adding ambient occlusion and making it toggleable required me to think carefully about how to make the scene look softer and less artificially sharp, while still keeping performance reasonable. Similarly, implementing multiple shadow resolution settings took a lot of testing and comparison, since each resolution needed to visibly affect the quality of the shadows while still running well enough for the user to interact with the scene.

A large amount of effort also went into building dynamic weather and shader based effects that were not part of a normal homework assignment. I added rain, wet ground, and fog so the entire environment could change depending on the user's settings, which meant I had to make sure these systems worked together with the lighting and time changes instead of conflicting with them. I also implemented custom shaders for the campfire and wildlife, which were important for giving the fire, smoke, sparks, and glowing fireflies a more natural and animated appearance than basic materials would allow. On top of that, I added several animated wildlife features, including Tung Tung Tung Sahur, the deer, and fireflies, along with aurora, stars, shooting stars, lanterns, flowers, and mushrooms to make the scene feel more alive. Overall, my contributions involved both technical rendering features and visual polish, and they required much more experimentation, debugging, and refinement than a standard homework assignment.

**Reflection (Sanjay):** In this project, I learnt how scenes with multiple light sources introduce a new level of complexity and depth compared to static scenes. I learnt that certain objects benefit from using certain types of lighting (such as a window for a cabin using a point light instead of an area light), while other objects may even need multiple different lighting types to achieve a realistic result (such as the campfire). In this project, I contributed to helping structure the overall scene (campground, forest ground, trees, and mountains), added multiple light sources in the scene (the sun, torches, campfire, windows, etc), and implemented dynamic lighting and shadows so that the intensities of certain light sources and shadows would not be static throughout the day and would dynamically change depending on the position of the sun/time of day. For what I would do differently next time, I would definitely consider adding some more light sources that added more structure to the scene (such as a firewatch tower), and maybe a random event that may happen in the forest (such as a forest fire) that would illuminate the scene differently.

**Reflection (Jayden):** In this project, I learned that making a scene look realistic is not just about adding more objects, but about making all the visual systems work together in a believable way. I especially learned how important shadows and shading are to the overall feel of a scene. Features like ambient occlusion, softer shadows, and different shadow map resolutions may seem like smaller details at first, but they have a huge impact on whether the environment looks flat or convincing. I also learnt that weather can significantly change how a scene is perceived, since effects like rain, fog, and wet ground do not just add atmosphere, but also affect how light interacts with the world. Working on these systems helped me better understand how rendering choices influence both realism and performance, and how much balancing is required when adding advanced visual effects.

In this project, I contributed many of the effects that made the scene feel more dynamic and stylized, especially during nighttime and bad weather conditions. I implemented the moon, aurora, stars, shooting stars, and multiple forms of wildlife, while also adding animation to wildlife so the environment did not feel static. I also worked on ambient occlusion, shadow resolution settings, dynamic weather, lantern lighting, smoke particles, and the custom shaders for the campfire and wildlife. Out of all of these, I think the shader work and atmosphere related systems taught me the most, because they required more creativity and more trial and error than simply adding geometry to the scene. If I were to improve the project further, I would probably spend more time making the transitions between weather states and time of day states even smoother, and I would also look into making the wildlife react more directly to changes in the environment, such as rain, darkness, or nearby light sources. Overall, this project helped me better understand how multiple rendering techniques can come together to create a much more immersive and polished scene.

**NOTE ON AI USE:** Since this project was very large, we used AI to help us brainstorm how to go about this project, mainly using Claude. The primary use of Claude was to help us understand what kinds of lighting to use for which kinds of objects, which features would make the scene a bit more realistic, and a lot of debugging. Before using code Claude provided, we tried it ourselves and tested how the implementation it recommended was different from our own.